

Dimensionless SG parameter result

Dimensionless geometric Savitzky-Golay experiment

Fixed reference scales

mass_kg: 2.3515975908123767
length_m: 0.19158499426207065
time_s: 0.13977207332379268
force_n: 23.061294513940194
torque_nm: 4.418197977129154
dimensionless_gravity: 1.0

Selected deterministic point (prior-free)

objective cost: 0.0008864918852971233
rotor delay [s]: 0.112521096281
gimbal delay [s]: 0.046667489993
optimizer success: True
optimizer message: three consecutive accepted steps satisfied relative objective-reduction tolerance
exact scale gauge: KKT hard constraint
unknown weak-mode cutoff: none
Jacobian rank: 13
Jacobian nullity: 1
common-scale analytic null norm: 2.022883282921508e-13
common-scale FD null norm: 1.093711701906961e-09

Dimensionless coordinate

log_mass_scale	-1.75705396387
log_second_moment_mode_xx	-5.38344331305
log_second_moment_mode_yy	3.76186426691
log_second_moment_mode_zz	11.1032331488
log_second_moment_mode_xy	1.10463311054
log_second_moment_mode_xz	-0.519583503965
log_second_moment_mode_yz	4.91194788065
dimensionless_cog_displacement_x	0.108068186189
dimensionless_cog_displacement_y	3.94375718495
dimensionless_cog_displacement_z	0.334254762005
log_force_effectiveness_1	-2.20133553469
log_force_effectiveness_2	-1.83898563113
log_force_effectiveness_3	-1.7171268454
log_force_effectiveness_4	-1.96715212758

Physical representative

mass [kg]: 0.40577395285096834
CoG body [m]: [0.01867953 0.75553417 0.07354795]
force effectiveness: [0.11065528 0.15897861 0.17958137 0.13985458]
inertia [kg m²]:
[2443.581473, 20.08975361, 4.376387316]
[20.08975361, 109.9279728, -505.993209]
[4.376387316, -505.993209, 2334.020868]
principal moments [kg m²]: [5.00346161e-02 2.44371534e+03 2.44376494e+03]

Per-bag raw residual-wrench RMS

failure_1_single: [8.88154961e-02 3.63994528e-02 1.16379502e-01 8.78207640e+02
3.34905828e+02 1.53461721e+03]